

Non-linear Simultaneous Equations – Solutions

LET – Mathematics and Statistics

1. *Solution:*

Since both equations are equal to y , we can use the substitution method by setting the right-hand sides equal to each other.

1. Set the Equations Equal

$$x^2 + 5x + 10 = 2x + 13$$

2. Rearrange into a Standard Quadratic Equation

$$x^2 + 5x - 2x + 10 - 13 = 0$$

$$x^2 + 3x - 3 = 0$$

This does not factor easily, so we must use the quadratic formula where $a = 1, b = 3, c = -3$.

3. Solve for x

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-3 \pm \sqrt{3^2 - 4(1)(-3)}}{2(1)}$$

$$x = \frac{-3 \pm \sqrt{9 + 12}}{2}$$

$$x = \frac{-3 \pm \sqrt{21}}{2}$$

4. Find the Corresponding y Values

Substitute each x value into the linear equation, $y = 2x + 13$.

For $x_1 = \frac{-3+\sqrt{21}}{2}$:

$$y_1 = 2 \left(\frac{-3 + \sqrt{21}}{2} \right) + 13 = -3 + \sqrt{21} + 13 = 10 + \sqrt{21}$$

For $x_2 = \frac{-3-\sqrt{21}}{2}$:

$$y_2 = 2 \left(\frac{-3 - \sqrt{21}}{2} \right) + 13 = -3 - \sqrt{21} + 13 = 10 - \sqrt{21}$$

Answer: The solutions are $(\frac{-3+\sqrt{21}}{2}, 10 + \sqrt{21})$ and $(\frac{-3-\sqrt{21}}{2}, 10 - \sqrt{21})$.

2. *Solution:*

This system is best solved using substitution. First, rearrange the linear equation to make y the subject.

1. Rearrange the Linear Equation

$$5x + y = 8$$

$$y = 8 - 5x$$

2. Set the Expressions for y Equal

$$8 - 5x = 20 - 4x - x^2$$

3. Rearrange into a Standard Quadratic Equation

$$x^2 - 4x + 5x + 8 - 20 = 0$$

$$x^2 + x - 12 = 0$$

4. Solve for x by Factorising

$$(x + 4)(x - 3) = 0$$

This gives two solutions for x : $x = -4$ and $x = 3$.

5. Find the Corresponding y Values

If $x = -4$: $y = 8 - 5(-4) = 8 + 20 = 28$.

If $x = 3$: $y = 8 - 5(3) = 8 - 15 = -7$.

Answer: The solutions are $(-4, 28)$ and $(3, -7)$.

3. Solution:

Use the substitution method. Rearrange the linear equation to find an expression for y .

1. Rearrange the Linear Equation

$$2x + y = 8 \implies y = 8 - 2x$$

2. Substitute y into the Second Equation

$$x(8 - 2x) + x^2 = 15$$

3. Simplify and Solve for x

$$8x - 2x^2 + x^2 = 15$$

$$8x - x^2 = 15$$

$$0 = x^2 - 8x + 15$$

$$0 = (x - 3)(x - 5)$$

This gives two solutions for x : $x = 3$ and $x = 5$.

4. Find the Corresponding y Values

If $x = 3$: $y = 8 - 2(3) = 8 - 6 = 2$.

If $x = 5$: $y = 8 - 2(5) = 8 - 10 = -2$.

Answer: The solutions are $(3, 2)$ and $(5, -2)$.

4. Solution:

This system can be solved efficiently using substitution.

1. Rearrange the First Equation

$$y + x^2 = 0 \implies y = -x^2$$

2. Substitute y into the Second Equation

$$x(-x^2) + x = 0$$

3. Simplify and Solve for x

$$-x^3 + x = 0$$

$$x(1 - x^2) = 0$$

$$x(1 - x)(1 + x) = 0$$

This gives three possible solutions for x : $x = 0$, $x = 1$, and $x = -1$.

4. Find the Corresponding y Values

If $x = 0$: $y = -(0)^2 = 0$.

If $x = 1$: $y = -(1)^2 = -1$.

If $x = -1$: $y = -(-1)^2 = -1$.

Answer: The three solutions are $(0, 0)$, $(1, -1)$, and $(-1, -1)$.